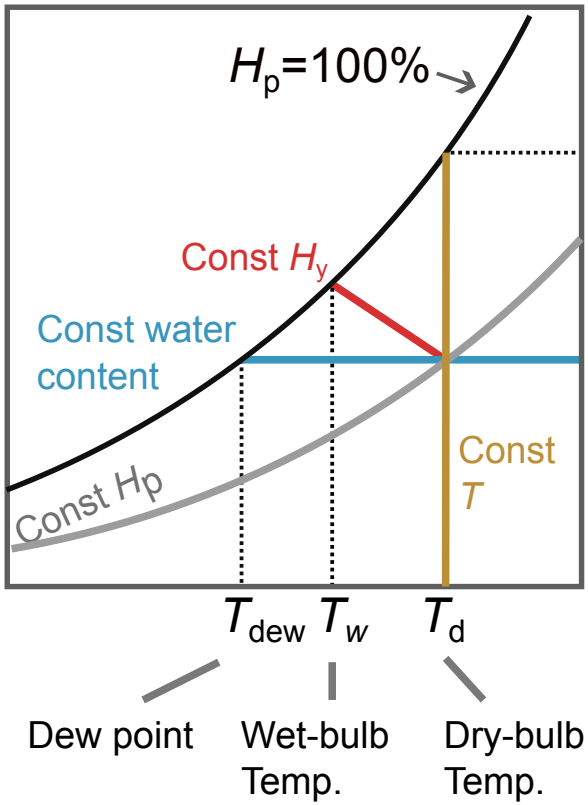
Chearsheet: Humidification & Concepts



H (humidity)

$$H_s = 0.622 \frac{p_{\text{vap}}}{p_T - p_{\text{vap}}}$$

$$H = \frac{M_A p_A}{M_B (p_T - p_A)} = 0.622 \frac{p_A}{p_T - p_A}$$

$$y = \frac{H/M_A}{1/M_B + H/M_A} \approx \frac{M_B}{M_A} H$$

Percentage humidity

$$H_p = \left(\frac{H}{H_s}\right) \times 100\%$$

Relative humidity (R.H.)

$$H_R = \left(\frac{p_A}{p_{\text{vap}}}\right) \times 100\%$$

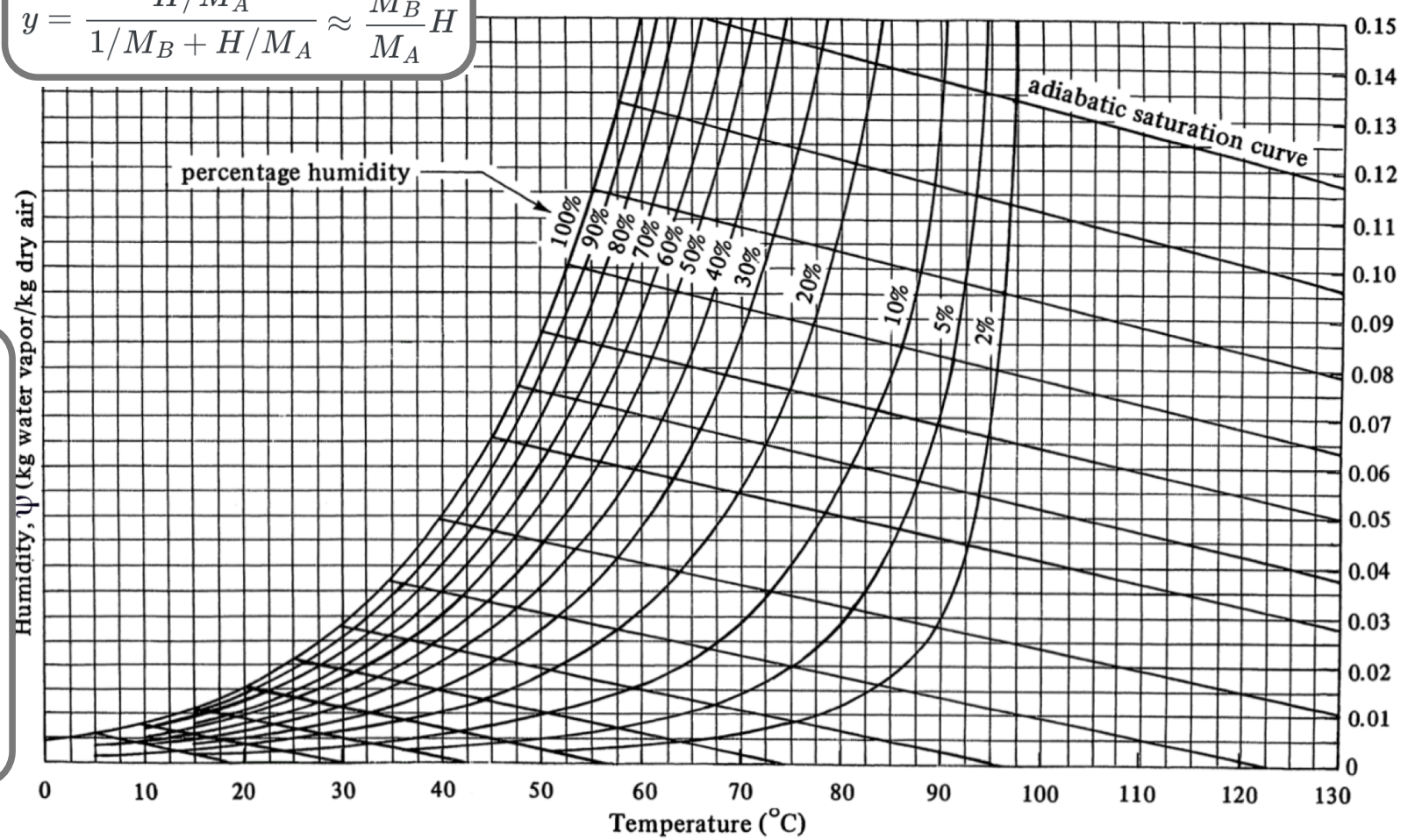
$$H_p = H_R \frac{p_T - p_{\text{vap}}}{p_T - p_A}$$

Humid volume (m^3)

$$v_H = \left(\frac{1}{28.97} + \frac{H}{18.02}\right) \frac{22.41}{273} T \left(\frac{1.013 \times 10^5}{p_T}\right)$$

Humid heat (kJ /kg dry air / K)

$$c_s = c_B + H c_A = 1.005 + 1.88H$$

Enthalpy(kJ /kg dry air)

$$H_y = c_s(T - T_0) + H\lambda_0 = (1.005 + 1.88H)T + 2501.4H$$